Igor Bryzhikov

Curriculum Vitae

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Research interests: zeroth-order and stochastic optimization, optimization under inexact oracles, memory-efficient LLM fine-tuning.

Education

2025 – 2029 (anticipated) **Moscow Institute of Physics and Technology**, *Phystech School of Applied Mathematics and Computer Science*, B.Sc. in Informatics and Computer Engineering

Relevant coursework:

Mathematics: Calculus, Multivariable Calculus, Analytic Geometry, Linear Algebra, Differential Equations, Probability Theory (in progress)

Computer Science: Algorithms and Data Structures, C++, Python (advanced course), Theory of Computation (in progress), Databases and SQL

Physics: Mechanics, Thermodynamics and Molecular Physics, Electricity and Magnetism, Analytical Mechanics

Average Grade: 8.7/10

Research Experience

Sep. 2026 – present **Research project**, *MIPT*, **Supervisor:** Prof. Alexander V. Gasnikov

Paper in preparation: *"Admissible Deterministic Noise in Zeroth-Order Convex Optimization under Markovian Orthogonality"*. Formalizing a Markovian model of deterministic noise in zeroth-order convex optimization and deriving two-sided bounds on its admissible level. Modelled floating-point rounding in low-precision MeZO fine-tuning as a lazy Markov chain and validated it experimentally.

Code and experiments: github.com/igor-rah/zero-order-optimization-research

Work Experience

Jul. 2026 – Oct. 2026 **BI Analyst Intern**, *T-Bank*, Data Security team

Automated the processes of a team that detects and masks sensitive data in the bank's data warehouse.

- Designed an efficiency metric for automated sensitive data-reports alert routing (operational load down 24%).
- Built two LLM tools: automated pre-review of Jira tasks (mentor review load down 2.5×) and a hint on masked fields for access requests (time to request down from 9 to 2.75 min).

Projects

Mar. 2026 – present **Training-data pipeline for LLMs**, *Data2Use startup*

Designed the pipeline stages for building LLM training datasets from unstructured corporate data; implemented validation, preprocessing and data-quality analytics (duplicates, gaps, schema errors) with unit tests.

Nov. 2025 – present **Conversational AI companion with a user profile**, *personal project*

Designed the database schema for model state and the rules for updating it; implemented the backend (Python, SQLite, SQLAlchemy, REST API) with background jobs; covered 78% of critical endpoints with tests.

Feb. 2025 – Apr. 2025 **Real-time threat detection in video streams**, *Russian National Entrepreneurship Olympiad, final*

Fine-tuned YOLOv8 for real-time threat detection in video streams (PyTorch).

Honors & Awards

2025 **Final of the Russian National Entrepreneurship Olympiad** (technology track), Finalist

Interregional Mathematics Olympiad for School Students (departmental universities), Prize-winner

Computer Skills

Python Jupyter, Data Analysis and Visualization (NumPy, Pandas, MatPlotLib), PyTorch, FastAPI

ML LLM, RAG, LoRA fine-tuning, prompt engineering, model evaluation

C++ Object-Oriented Programming, Algorithms and Data Structures, Metaprogramming

Other LaTeX, Git, Bash, Docker, CI/CD, Apache Superset

Languages

Russian (Native), **English** (B2 – Upper-Intermediate)

Hobbies

Music Completed a ten-year piano programme at a music school

Sports Member of the MIPT Boxing Team

Other Outreach to former MIPT faculty for the Teacher Gratitude Platform (MIPT & Phystech Union); member of the "Generation" mentorship programme led by Viktor Kuznetsov (founder of VseInstrumenti.ru); volunteering at summer festivals